



Tiga: Compiling Graph Message Passing at Scale

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Abstract

Graph message passing offers a common way to express learning algorithms, physical simulations, and numerical solvers. Efficient execution depends on interaction structure and data movement, which can be obscured when a program is expressed as a sequence of tensor operations. On memory-constrained systems such as laptops, materializing connectivity and intermediate messages can also exhaust device memory. We present Tiga, a just-in-time compiler that separates the definition of a message-passing program from how its interactions are traversed, computed, and stored. Tiga preserves graph relations and reducer algebra in a multi-level intermediate representation, enabling traversal specialization, fusion of relation generation with aggregation, and reverse-mode automatic differentiation. Backend-specific lowering targets CPUs and GPUs. Its runtime extends execution beyond device-memory capacity by streaming graph partitions from disk through host memory with page-sized device staging buffers; partition ownership and halo exchange extend the same programming model to distributed execution. A Python interface interoperates with ordinary PyTorch tensors and autograd for forward and backward computation. Numerical checks validate outputs and gradients for differentiable workloads. A differentiated geometric workload characterizes the memory–time tradeoff of fused forward and backward execution. Evaluation against matched Torch and PyTorch Geometric baselines demonstrates reduced runtime and device-memory use for generated-relation workloads, while offloaded forward execution processes billion-edge graphs on a single memory-limited GPU. Measurements on heterogeneous devices further characterize the communication and load-balance costs of distributed execution.

1 Introduction

Message passing can be a general programming model, not merely an interface for graph neural networks. Computation is organized as entities with state, relations that determine which entities interact, local message functions, and successive state updates. The graph need not be the input problem: it can represent the structure of the computation itself. Tensor contractions, convolutions, particle interactions, iterative solvers, attention and graph algorithms can all be viewed through this lens. General message-passing models have a long history, including the actor formalism [9].

We present Tiga (*Target-Independent Graph Acceleration*), a just-in-time (JIT) compiler for graph message-passing programs. Target independence refers to the separation of program semantics from hardware-specific lowering: a message function describes an interaction, while traversal, kernel mapping and data placement determine its execution. The implementation targets CPU and NVIDIA GPU backends, with an extensible backend integration interface for additional targets.

The long-term goal is a common programming substrate for dense, sparse and dynamically generated computation, rather than a collection of unrelated operator-specific interfaces. Relations describe interaction, messages describe local computation, reducers describe combination, and composition with control flow describes how state evolves. A compiler can then choose how these programs execute across kernels, memory tiers and devices while preserving their meaning. Generality concerns what can be expressed; it does not imply that every problem admits an efficient parallel encoding or that every encoding is supported by the current implementation. Section 2.1 makes this distinction precise.

This common form does not imply a common optimal implementation. A relation in compressed sparse row (CSR) storage needs an indexed traversal; a dense relation may need no edge array; a generated radius relation may profit from a spatial directory. Feature width, degree skew, gradient requirements and data residency change the suitable plan.

Three practical costs motivate this design. First, explicitly expanding a radius or nearest-neighbor relation can create indices, geometry and messages that are consumed only once. Second, a graph larger than GPU memory requires bounded storage movement even when its arithmetic is simple. Third, dividing a graph across devices introduces halo traffic and rank imbalance; doubling the GPU count need not reduce completion time.

The central idea in TIGA is to separate logical interactions from their physical execution. Keeping relations visible to the compiler allows generated neighbors to be consumed directly by aggregation instead of first being stored as a full edge list. Separating logical values from their residence allows a large graph to be processed in pages, while ownership and halo exchange define its distributed execution. Backend-specific lowering then maps the computation to CPU or GPU kernels. These are complementary ways to control the cost of the same program, rather than separate application models. For memory-constrained systems such as laptops, paging is particularly important: graph capacity need not be limited to accelerator memory. Current executable backends and extension boundaries are distinguished in Table 4.

TIGA retains relations and reducer algebra as compiler-visible information. It is not a model library, optimizer package or replacement for PyTorch. Ordinary applications retain `torch.Tensor` and PyTorch autograd [17]. Reverse-mode automatic differentiation computes vector-Jacobian products (VJPs), propagating an output loss gradient back to input fields and model parameters [2]. Torch is not a base installation dependency: a standalone native path supplies CPU computation, message passing and gradients without it. Native tensors also expose compiler/storage functionality not uniformly available through Torch. This split is an explicit interoperability boundary, not a claim that both interfaces support all operations identically.

The report makes the following contributions:

- A relation-aware compiler design that carries field roles, reducer algebra and derivative requirements from the programming interface to traversal and kernel selection.
- A runtime model that separates logical values from storage residence and rank ownership, enabling paged and distributed execution with explicit data movement contracts.
- An evaluation of runtime and memory efficiency, billion-edge capacity, and heterogeneous distributed execution, accompanied by numerical checks, reproducible measurements, and a differentiable EdgeNN example.

The contribution is the integration of these mechanisms around a shared message-passing representation; Section 2 relates the individual ideas to prior work. The experiments compare time and allocation with matched Torch/PyG paths, execute one billion explicit edges on a 16 GiB GPU, and measure the cost of two-host NCCL graph partitioning. The results show that generated radius traversal and exact kNN improve latency and allocation on the measured workloads; paging extends executable capacity; and the current equal-row two-GPU schedule is slower than the faster GPU alone. This connects each mechanism to a measured benefit or cost rather than assuming that one execution strategy is efficient for every relation and interconnect.

Reading and using the system. Application development starts with the programming model in Section 3 and the complete forward/backward example in Appendix A; Appendix B expresses causal attention with the same graph, edge and reducer interface. Compiler and runtime development is covered in Sections 4–6; reproduction details appear in Section 7. The implementation, API documentation, and report artifacts are available separately so that implementation details and experiments can be inspected alongside the design.

2 Background: why graph-aware compilation?

2.1 Message passing as a general computational model

Message passing is broader than neighborhood aggregation in a fixed-depth graph neural network (GNN). In the general model, stateful entities perform local computation, communicate values or control information, and continue through repeated transitions. The actor formalism treats communication as an organizing principle for data, procedures and control [9]. Pregel gives a graph-oriented, iterative realization through vertex programs and messages [16]. Tiga uses structured relations and reducers rather than implementing arbitrary actor mailboxes or adopting Pregel’s execution protocol. In graph learning, Message Passing Neural Networks (MPNNs) formalize learned messages and aggregation across repeated node updates [8]. Tiga uses this local-computation interface as a compiler input, while also supporting non-neural message functions such as stencil updates and weighted sums.

Universality and its conditions. With sufficiently expressive local transitions, repeatable control and arbitrarily extensible state, a message-passing model can in principle express any computable algorithm. A constructive way to see this is to encode a Turing machine’s tape as a chain of entities. Each entity stores a tape symbol; a message carries the head’s finite control state. Receiving the head message updates the symbol and sends the next head state to the left or right neighbor. Allowing the chain and the number of transitions to grow reproduces the machine’s computation. This is an expressiveness argument for the general model, not a universality proof for Tiga’s present bounded loops, fixed-shape tensors or finite-memory backends. It refers to computable problems, not undecidable ones, and says nothing by itself about time, memory or useful parallelism.

A common language for numerical programs. The practical opportunity is to retain a useful computational structure without requiring every application to start from an explicit graph dataset. Elementwise maps use independent entities or self-relations. Matrix multiplication connects each output row to contributing source rows, with weighted vector messages and a sum reducer. Convolution and finite-difference stencils use regular local relations; particles use coordinate-dependent radius or kNN relations; attention uses dense or masked relations and a normalization-aware reducer. Repeated message passing with state updates expresses time stepping, iterative solvers and graph traversals. These are representations within a shared conceptual model, not a claim that every specialized operation has an equally optimized Tiga path. The language ambition is to make such programs composable and their interactions compiler-visible; the implementation and evaluation below establish concrete slices of that ambition.

Why a compiler is needed. An edge function can be only a multiply, yet its execution may require reading an index, gathering a feature vector, creating an edge-sized temporary and scattering into a destination. For E edges and feature width F , explicitly materializing messages requires storage proportional to EF , even when the final output has only NF values. A fused traversal can instead accumulate messages directly. Fusion is not universally optimal: a library sparse matrix operation, feature tiling, recomputation or a different graph layout may win depending on shape, degree distribution and reuse. Consequently the relevant compiler question is

which semantic information remains available when each decision is made, not whether a system uses the phrase “message passing.”

PyTorch Geometric supplies a graph-learning programming model with gather–scatter and optimized sparse execution paths [6, 7]. Graphiler makes the compiler connection explicit: its message-passing data-flow graph annotates graph-related tensor residency and movement, enabling reductions in redundant computation and intermediate storage [22]. Tiga likewise retains source-, edge- and destination-associated semantics, and carries these roles into relation traversal and differentiation.

2.2 Graph and feature schedules, fusion, and training

FeatGraph composes sparse traversal templates with feature-dimension UDFs and schedules [10]. It shows why graph traversal and within-feature computation should be optimized together. Seastar offers vertex-centric Python programming and generates fused forward and backward GPU kernels [21]. Tiga shares the goal of fusing local computations, and combines it with explicit relation constructors and storage/ownership representations.

GALA is especially relevant to a multi-level design. It separates data and compute representations, composes intra-operator and inter-operator transformations, and reasons about training-specific optimization opportunities across forward and backward computations [15]. Tiga separates relation, traversal and execution representations; its current evaluation focuses on forward workloads, with path-specific gradient implementations described in the differentiation section. Whole-training-program optimization remains beyond that evaluated scope.

GraphIt separates the graph algorithm from scheduling choices [24]. That distinction helps explain why a relation-level operation should not prematurely encode threads, blocks or traversal order. Tiga additionally seeks to preserve numerical field operations, reducer state and differentiation requirements alongside traversal scheduling.

2.3 Relations and sparse iteration beyond GNNs

Ebb represents geometric domains through a relational data model and separates simulation code, domain libraries and parallel runtime implementation [3]. Simit connects hypergraph-local computation to global tensor algebra for physical simulation [12]. They are important precedents for a graph language that is not limited to neural networks. Tiga’s particle, solver and stencil examples belong in this lineage of relational numerical programming.

TACO derives compound dense/sparse tensor kernels from high-level algebra using iteration and merge structures [11]. Its format-abstraction work further separates logical computation from sparse storage levels [4]. SparseTIR develops composable sparse formats and transformations for deep-learning operators [23]. These works motivate keeping coordinate traversal distinct from physical layout and execution mapping. Tiga’s Iter IR preserves relation-coordinate structure for its supported traversal families; the implementation does not provide a general sparse-format algebra.

Triton supplies a lower-level tiled compilation interface [20], while MLIR supplies infrastructure for composing dialects and lowering passes [14]. Tiga uses both: graph-aware representations are upstream of a current NVIDIA provider. The provider performs device-code compilation after Tiga’s graph-aware transformations.

2.4 Capacity: running the graph at all

Reducing execution latency and reducing the resident working set are related but distinct goals. If a graph does not fit a resource budget, a faster in-memory kernel is insufficient. GraphChi established disk-based large-graph computation on a single machine through partitioned processing and sliding windows [13]. X-Stream uses edge-centric streaming partitions to favor sequential

access over random graph-data access [18]. These systems are relevant even though they do not share Tiga’s differentiable Tensor interface.

Legion separates logical regions and task privileges from placement and physical instances, providing a model for reasoning about locality, independence and data movement [1]. Tiga’s storage versions, task dependencies and halo ownership apply this separation to the graph runtime described in Section 6.

These systems distinguish executing faster from making execution possible. Tiga evaluates both, plus the cost of distributing the same graph. The capacity experiment traverses explicit disk-backed CSR through host staging to CUDA; the distributed experiment measures rank ownership and halo movement separately. Neither a transfer-bandwidth microbenchmark nor a successful communicator setup alone answers these application-level questions.

FlashAttention is a related example of IO-aware exact computation [5]: avoiding an intermediate can matter as much as reducing arithmetic. Tiga’s dense relations and streaming reducers expose similar scheduling opportunities, while optional tile pruning changes the mathematical contract and is distinct from exact attention. The experiments in this report concern graph aggregation and do not claim attention-library parity.

Prior work	Relevant mechanism	Connection to Tiga
Graphiler [22]	Graph-aware dataflow/residency	Source, edge and destination field roles
FeatGraph [10]	Sparse traversal + feature schedules	Joint graph/feature traversal selection
Seastar [21]	Vertex-centric fused forward/backward	Message fusion and derivative generation
GALA [15]	Composed data/compute and training transforms	Layered representation and transformation scope
Ebb / Simit [3, 12]	Relational simulation / hypergraph algebra	Numerical programs beyond GNNs
GraphIt [24]	Algorithm/schedule separation	Separate meaning from execution mapping
TACO / SparseTIR [11, 23]	Sparse iteration, formats, transformations	Relation traversal separate from storage
GraphChi / X-Stream [13, 18]	Disk-oriented partitioned execution	Bounded graph pages and storage traffic
Legion [1]	Logical regions and physical placement	Storage instances, ownership and dependencies

Table 1: Prior work and its relationship to Tiga’s design.

Scope of the distinction. Tiga does not introduce message passing, kernel fusion, sparse traversal or offloading individually. Its design contribution is their integration around explicit relation semantics: a generated relation can feed a reduction without a stored edge list, while a stored relation can retain the same local program under a bounded-residency runtime. Field roles and reducer contracts constrain legal transformations, and differentiation is attached to the supported execution route. The distinction from a library of operators is the retained program representation; the distinction from existing compilers is the particular combination of relation constructors and storage/ownership contracts, not a claim of uniformly broader coverage or faster execution. Measurements below compare implemented paths, not every cited system.

3 Programming model

3.1 Relations, fields and messages

Let source entities and destination entities be distinct finite sets. A relation is an ordered collection of edges with source and destination maps. Duplicate edges remain separate messages. For a destination, the computation is

$$m_e = f(x_{s(e)}, z_{d(e)}, a_e; \theta), \quad y_i = g\left(z_i, \text{finalize}\left(\bigoplus_{e:d(e)=i} \text{lift}(m_e)\right); \theta\right).$$

Here source fields, destination fields, edge attributes and shared parameters are denoted by x , z , a and θ , respectively. The edge function f forms a message; g optionally updates destination state from the aggregate. The reducer supplies its identity, lift, combine and finalize behavior. An empty row starts from the identity; finalization determines the resulting empty-row convention. This convention is reducer-specific. Declaring a reducer does not define which edge-local values enter it: the edge function does. Simple sum messages return a tensor directly. A structured streaming reducer such as online softmax currently packages a score and value using an explicit reducer call inside the edge function. That call does not execute an aggregation once per edge; it binds the message inputs for capture.

CSR is destination-oriented: row boundaries enumerate destinations and column indices identify sources. The example below has two edges into destination 0, one into destination 1, and none into destination 2. A self edge is ordinary; there is no automatic reverse edge.

```
import torch
import tiga as tg

class WeightedSum(tg.MessagePassing):
    reducer = tg.sum()
    def edge(self, src, dst, edge):
        return src.x * edge.w

graph = tg.Graph.from_csr(
    torch.tensor([0, 2, 3, 3]), torch.tensor([0, 1, 1]),
    num_src=2, validate="full")
x = torch.tensor([2., 3.], requires_grad=True)
w = torch.tensor([4., 5., 2.], requires_grad=True)
y = WeightedSum()(graph=graph, src={"x": x}, dst={}, edge={"w": w})
dx, dw = torch.autograd.grad(y.sum(), (x, w))
# y=[23, 6, 0], dx=[4, 7], dw=[2, 3, 3]
```

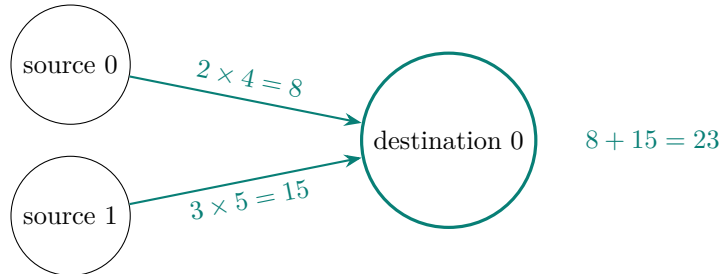


Figure 1: One destination row: two edge messages become one result. Other destinations are intentionally omitted.

3.2 Composition and topology

Dense and triangular constructors describe relation structure, while `Graph.cat` forms disjoint blocks by offsetting source and destination IDs independently. For example, concatenating triangular graphs of sizes two and three represents independent causal sequences. It introduces no cross-block edges and currently materializes CSR; generic composition is not yet an implicit-storage optimization. Cumulative sequence boundaries remain a compatibility constructor and must start at zero.

Radius and nearest-neighbor builders depend on coordinates. Their relation realization may differ by execution path. Topology selection is discrete; the selected graph is a fixed snapshot for derivative semantics. In-place mutation of captured indices is not a supported topology-update mechanism. Transpose and composition can reorder edges, so explicit edge fields require matching reordering. A constructor accepting an input does not certify every consumer, provider or gradient combination.

4 Compiler organization

Tiga uses the extensible Multi-Level Intermediate Representation (MLIR) infrastructure [14]. An intermediate representation (IR) records a program in a form that compiler passes can analyze and transform. Tiga’s representations separate the mathematical computation from traversal and execution decisions. Applications use the Python interface; the `gf.*` dialect operations below are its compiler-facing representation. Domain, Iter, Kernel and Task/Storage name semantic layers, each containing several operations.

Layer	Example operation	Preserved decision
Domain	<code>gf.apply</code>	Relation, field roles, message regions and reducer algebra
Iter	<code>gf_iter.traverse</code>	Traversal organization and reduction structure
Kernel	<code>gf_kernel.launch</code>	Work mapping and local resource organization
Task/Storage	<code>gf_task.launch</code>	Dependency, transfer and readiness ordering
Tensor	<code>gf_tensor.mul</code>	Tensor expressions, views and gradient computation

Table 2: Compiler representations and the decisions they preserve. Execution paths use the layers needed by the selected operation and backend.

4.1 Design contract: delay decisions, retain legality

The layers answer different questions about the same program. Domain asks what messages and reductions mean. Iter asks which relation coordinates must be visited. Kernel asks how that work is mapped to execution resources. Task and Storage ask when work may run and which physical copy supplies its operands. Tensor IR is a numerical-expression layer used alongside these decisions, not necessarily a final fifth stage. The split avoids encoding a GPU schedule in the mathematical interface, but does not make scheduling automatic or optimal.

The key invariant is preservation of the observable outputs and supported VJPs for the declared input contract. Shape, dtype, device, field role, reducer state, topology snapshot and data-access effects are parts of that contract. A pass that cannot establish a required property must reject the transformation or select an explicitly diagnosed alternative. A topology version in the IR is a validity annotation; it does not make arbitrary external in-place mutation safe.

4.2 Domain: field roles and reducer algebra

In the example, the source field is indexed by a source ID and the edge weight by an edge position. Both are floating-point vectors, but exchanging their roles changes the program. `gf.apply`

therefore retains `input_roles`, `input_names`, reducer symbols, region kinds and input segments. Its region contains the edge-local arithmetic, not an already materialized global message array. A relation separately retains cardinalities, origin, lifecycle, version and the available degree constraints.

A reducer describes identity, lift, combine and finalize regions, including the types of its intermediate state. This is richer than attaching the string “sum” to an arbitrary scatter. A product or online-softmax reducer needs different state and different differentiation rules. Empty rows start from the identity and still obey finalize. Associativity admits regrouping; commutativity admits reordering. These declarations are semantic obligations, not properties the compiler proves for unrestricted user-written Python. Floating-point sums require tolerance-aware validation when their reduction tree changes.

The full fixture and four unedited compiler outputs are included in `generated/ir/`. This excerpt shows the fixture’s edge region; its enclosing Domain operation is omitted:

```
^bb0(%source: f32, %edge: f32):
  %message = arith.mulf %source, %edge : f32
  "gf.yield"(%message) : (f32) -> ()
```

The block arguments represent one logical edge invocation. The region itself does not say which CPU core, CUDA thread or memory tier executes it.

This representation makes the materialization decision explicit. An eager implementation gathers a source row for every edge, writes its weighted message and then reduces those messages. A fused row traversal reads the same source and weight, forms a tile-local message and immediately updates reducer state. For feature width F , the former can allocate an edge-message array with EF entries; the latter needs only tile state in addition to graph, inputs and the NF output. The edge set and message arithmetic are unchanged. The fixed-topology study in Section 7.5 holds that edge set constant to examine execution-path differences. Generated radius traversal additionally avoids accepted-edge storage, but that distinct benefit is not isolated by a stored-CSR comparison.

4.3 Iter: coordinates before threads

The Domain-to-Iter pass replaces the application with `gf_iter.traverse` and preserves the reducer and edge region. In the checked CSR fixture, the attributes `coordinate_hierarchy = "compressed-row"` and `ordering = "destination-major"` describe a row-oriented coordinate walk. These names do not mean that the operation is already an explicit sequential loop, or that all threads use the same row length.

For the three-row example, `row_ptr=[0,2,3,3]` determines edge ranges $[0, 2)$, $[2, 3)$ and $[3, 3)$. The edge position indexes the weight; `col_idx[e]` supplies the source ID. Thus the first row visits source IDs 0 and 1, forms messages 8 and 15 and produces 23. The second produces 6; the empty third row produces the sum identity 0. Confusing edge positions with source IDs would already invalidate lowering before any hardware schedule.

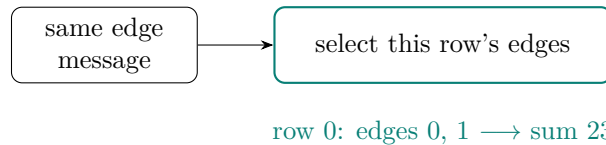


Figure 2: Iter introduces coordinate traversal, not a thread assignment. The highlighted decision is row membership.

Dense and generated relations need different coordinate producers. Materializing all edges merely to resemble CSR can destroy the reason to retain an implicit relation. Conversely, pre-

serving a constructor in the frontend does not prove that every backend consumes it implicitly: graph concatenation currently materializes CSR. Format abstraction and schedule selection must be inspected on the path actually chosen.

4.4 Kernel: a concrete, guarded schedule

The fixture’s Kernel operation consumes CSR buffers and field tensors. Its selected bounded-ragged schedule records these attributes:

```
block_rows = 16, block_neighbors = 16,  
num_warps = 4, pipeline_stages = 1
```

These are values for this fixture, not global Tiga defaults. Its execution roles distinguish work-group destination rows, subgroup neighbor reduction and scalar lanes. Resource and handoff attributes describe global reads, private register state, subgroup reduction and global output writes.

The important transition is from enumerating the correct coordinates to deciding how multiple coordinates share hardware. A maximum-degree guard can admit a masked bounded-neighbor tile; without that bound, the same launch geometry may be incorrect or wasteful. A high-degree split requires explicit partial states and a legal merge. No universal schedule should be inferred from the simple case’s kernel. Provider-neutral scheduling still needs target capabilities and resource constraints; it does not imply identical machine instructions on every backend.

4.5 Task and Storage: dependencies are not residency

The checked distributed fixture introduces partition and halo planning, then pack, exchange and unpack events. Interior work depends on partition readiness; boundary work additionally depends on unpack completion. A join represents the completion of both branches. This fixture demonstrates the dependency vocabulary; the public executor uses the communication-then-compute order described in Section 6.

Storage represents a logical region and version separately from physical instances, which carry capacity, memory space, device and layout. `gf_storage.transfer` produces a readiness event; `gf_storage.release` depends on completion before storage is recycled; `gf_storage.join` joins events without specifying a host-wide synchronize. These operations establish a vocabulary for movement and lifetime, not proof that a cost model automatically supplies an out-of-core schedule for any kernel.

For a paged graph, correctness requires that the page’s topology and field version agree, transfer completes before consumption, and storage is not reused until the last consumer finishes. The current public CPU/CUDA paged-CSR executor uses host-side orchestration for these dependencies. Whole-Tensor least-recently-used (LRU) eviction is a separate policy from partitioning one oversized active working set.

4.6 Transformations, verification and differentiation

Differentiation can change the access pattern: source gradients accumulate over reverse incidence, and edge gradients retain edge identity. A forward schedule need not be a good backward schedule. The compiler therefore needs a distinction between semantic VJP generation, schedule selection and runtime saved-state management. For current Torch CSR replay, that semantic bridge is implemented through Torch autograd rather than a generated TTIR backward kernel; dedicated edge-NN and native VJP paths have different contracts.

From the report repository, reproduce the architectural example with an installed `gf-opt` compiler executable:

```
gf-opt generated/ir/distributed-plan.mlir \
```

Transformation	Required fact	Failure if ignored
Row regrouping	Reducer association/order contract	Different state or numerical behavior
Bounded neighbor tile	Degree/shape guards and masked accesses	Truncated rows or out-of-range loads
Producer-consumer fusion	Compatible effects and dependencies	Stale values or duplicated side effects
Topology reuse	Matching frozen snapshot and bindings	Wrong edges after mutation
Storage reuse	Last consumer completed	Use-after-recycle or stale version
VJP construction	Saved/recomputed primal state and fixed topology	Incorrect input/parameter gradients

Table 3: Legality obligations, not a claim that every possible violation is statically proven absent. Runtime guards and negative tests remain necessary.

```
-gf-lower-domain-to-iter \
-gf-lower-iter-to-kernel -gf-select-kernel-schedule \
-gf-plan-distributed-tasks
```

The `generated/ir/` directory contains the individual stages in `domain.mlir`, `iter.mlir`, `kernel.mlir` and `task.mlir`. The artifact exporter checks them against compiler output before copying them.

4.7 Implemented routes through the layers

The source example `generated/ir/distributed-plan.mlir` contains explicit placement metadata and follows the Domain, Iter, Kernel and Task progression. It is supplied as a compiler-level input so that each lowering decision can be reproduced independently of Python capture.

Native Tensor programs, including native CSR message passing, can lower through Tensor IR to Vector/SCF/MemRef and LLVM execution. Relation-oriented CUDA paths retain graph structure through Iter and Kernel representations. Task and Storage operations appear where needed for dependencies and movement; a simple call does not necessarily materialize each architectural layer as an inspectable artifact.

4.8 Providers and inspectability

A backend is an execution target and its lowering route; a provider is the integration component that invokes a target toolchain and exposes its executable to the runtime. A transport moves data between ranks and does not generate compute kernels.

The NVIDIA provider hands serialized Triton IR (TTIR) to Triton [20], which continues to device code. Serialization avoids requiring the provider and Tiga to share one in-process MLIR ABI. Target capability information still influences earlier scheduling. CPU execution uses LLVM and an ExecutionEngine binding. Table 4 distinguishes executable routes from extension interfaces.

The application can inspect `program.explain()` and available stage/code artifacts. Native tensors additionally expose execution metadata after realization. Available artifacts depend on the selected provider. Small native expressions under an automatic policy may use a Python reference evaluator. Strict native validation explicitly requires the native policy and checks the reported backend.

For a new target, the implementation work lies in capability checks, legal lowering, provider compilation/loading, and runtime buffer and event handling. The common IR preserves the computation, while those components account for target-specific layouts and resource limits. The

Backend / target	Current executable route	Evidence and boundary
CPU / LLVM	Tensor IR through Vector/SCF/MemRef and LLVM ExecutionEngine	Native CPU forward/VJP and paged CSR; supported reducer/layout subset
NVIDIA / CUDA	Relation/Kernel IR to serialized TTIR, vendor Triton, PTX/cubin; CUDA Driver runtime	Torch/PyG comparisons, billion-edge paged forward and two-host NCCL execution; backward remains path-dependent
Torch adapter	Ordinary Torch tensors and autograd; dispatch to compiled or library/reference routes	Interoperability layer, not an independent hardware backend; some CSR VJPs replay Torch
AMD / Hygon	Provider ABI and conformance scaffolding	Pending provider integration and real-hardware validation; no speedup reported
Apple Metal	Provider interface only	Pending legal lowering and device validation; no speedup reported
Other accelerators	Extension points	Device implementations require provider integration and hardware validation

Table 4: Backend execution paths. CUDA graph paging uses host staging, and MPI/TCP/NCCL are transports rather than kernel code generators.

provider interface and development guide provide implementation entry points. Forward and VJP conformance must be checked separately; adding a device name alone does not supply either executable path.

4.9 Generated relations and executable reuse

Generated relations allow a consumer to evaluate candidate neighbors and accumulate messages without storing the accepted edges or messages globally. For low-dimensional Euclidean radius relations, a cell directory restricts candidate traversal to nearby spatial cells. An exact distance predicate then selects the accepted neighbors. The benefit depends on cell occupancy and geometry; a custom distance function requires a compatible spatial bound before this traversal is legal. Section 7.2 evaluates a radius workload that uses this build-and-consume strategy.

For generated kNN, one program scans candidate tiles for a destination query, retains a power-of-two padded top-k state, and consumes selected source features without materializing adjacency. A candidate’s ordering key contains its squared distance and source index, so distance ties have a stable order. Before sorting a tile, the kernel compares its minimum key with the largest retained key. If the tile cannot improve the retained state, sorting and merging are skipped exactly; distance evaluation is not skipped. Otherwise, local sorting and the retained-state merge execute normally. For non-power-of-two k, the padded state makes the test conservative. The search still examines all query–candidate pairs and is not a spatial-index algorithm.

Repeated public kNN calls can bind coordinates from a new Graph descriptor with equivalent relation semantics. The executable guard checks query/source counts, k, self-exclusion and domain identity semantics, coordinate and field shape/dtype/device/stride, field names, and the absence of gradient-bearing inputs. Coordinates and feature values remain live operands, not cached graph results. This separates changing data from changing programs and removes repeated capture/lowering/provider preparation for compatible dynamic calls.

4.10 Programs and bounded control

Straight-line message-passing calls can be captured as a typed acyclic `GraphProgram`. Independent legal leaves may fuse horizontally; dependencies remain ordered. `@tg.jit` also captures

supported Python loop forms, while `repeat` and `while_loop` offer explicit structured control over native values. Loop-carried shape, dtype and device are invariant. Bounded while requires a scalar boolean condition and a finite maximum iteration count. CPU lowering supports the implemented structured loop slice; general multi-state CUDA loop plans and structured reverse-loop lowering remain incomplete.

5 Differentiation and numerical contracts

Forward execution computes outputs from input fields and parameters. Reverse-mode automatic differentiation propagates an output cotangent back through that computation [2]. For a program with Jacobian J_F , the VJP is

$$y = F(x), \quad \bar{x} = J_F(x)^\top \bar{y}.$$

The bar denotes a loss derivative: for a scalar loss, the output cotangent is the derivative of that loss with respect to y . The full Jacobian need not be materialized. Tiga implements this operation for supported native computations and integrates differentiable Torch-facing calls with PyTorch autograd.

For fixed relation indices, weighted sum has a simple adjoint. With output cotangent denoted by a bar, source and edge derivatives are

$$\bar{x}_j = \sum_{e:s(e)=j} w_e \bar{y}_{d(e)}, \quad \bar{w}_e = x_{s(e)} \bar{y}_{d(e)}.$$

Repeated source indices require accumulation, not assignment. Empty destination rows contribute no source/edge derivative. The example in Section 3 provides an independently checkable forward and backward value contract. For its summed-output loss, the output cotangents are all one. A source that appears on multiple edges receives the sum of their weighted contributions; each edge weight receives the feature value from its own source.

5.1 Native and Torch paths

Native reverse-mode transforms construct derivative expressions from supported Tensor and relation operations. Torch-facing calls preserve ordinary Torch outputs and autograd integration, but their backward implementation is path-specific. Some compiled CSR forwards replay fixed-topology Torch semantics during backward. That route is differentiable without being a compiled TTIR backward kernel. Explicitly wrapping a Torch tensor into a native tensor shares storage but does not merge the two autograd histories.

Supported traced edge neural networks have a CUDA tile forward and a symbolic-VJP recompute kernel. The tests compare output and gradients of source features, positions and every module parameter to an eager reference. CPU tests additionally exercise explicit edge attributes with double-precision finite differences. These checks establish the tested first-order envelope, not arbitrary Python capture or universal higher-order differentiation. ReLU-like nondifferentiabilities need appropriate test points, rather than unqualified finite-difference equality at a kink.

For application code, set `requires_grad` before evaluation and call `torch.autograd.grad` or `loss.backward()` on ordinary Torch outputs. Native tensors use `tg.autograd.grad`. Gradients are taken with respect to values on the selected topology: radius cutoff membership and kNN index selection are discrete and are not themselves differentiated. The EdgeNN example in Appendix A checks feature, position and parameter gradients against an explicit reference. The billion-edge capacity measurement uses CUDA paging and therefore establishes forward capacity, not billion-edge training capacity.

Execution path	Forward	Backward
Torch CSR	Compiled or adapter execution	Torch autograd; some routes replay fixed-topology Torch operations
CUDA EdgeNN	Fused edge evaluation and reduction	Symbolic VJP with activation recomputation; input, position and parameter gradients
CUDA streaming attention	Implicit dense or triangular relation; tiled online softmax	No streaming backward; explicit Torch reference supports semantic gradient checks
Native CPU CSR and paging	Supported native operations and paged reducers	Native VJP, including supported disk-backed fields
CUDA paged CSR	FP32 paged execution with resident output	Not implemented on this path
Generated exact kNN	Fused selection and aggregation	No generated backward on this optimized path
Distributed native programs	Local computation after halo exchange	Local VJP plus reverse halo accumulation on supported CPU/CUDA paths

Table 5: Differentiation follows the selected execution path, not just the graph constructor. These are capability boundaries; the primary performance experiments measure forward execution, while Section 7.5 adds a differentiated EdgeNN workload.

5.2 Online softmax and approximation

For messages consisting of a score and value, the streaming state tracks the maximum score, normalized mass and weighted value accumulator. Two nonempty states combine by rescaling into a common maximum:

$$m = \max(m_a, m_b), \quad \ell = e^{m_a - m} \ell_a + e^{m_b - m} \ell_b, \quad o = e^{m_a - m} o_a + e^{m_b - m} o_b.$$

Finalization divides the accumulator by the mass for nonempty rows; empty-state handling must avoid undefined subtraction of infinities and follow the implementation’s reducer convention. This numerical structure is related to IO-aware attention algorithms [5]; it does not imply that every Tiga path achieves their performance or full feature coverage. Appendix B gives an executable causal-attention example and separately checks its compiled forward and Torch semantic gradients.

An optional positive pruning threshold enables an approximate dense-attention forward path. Scores and key reads are still needed for the pruning decision; eligible tiles skip value loading and accumulation. A large score gap can make a tile’s contribution small, but a block-level criterion is not a per-query error bound. Threshold choice affects accuracy and work saved, and may not improve latency. The exact default is a disabled threshold, not a threshold of zero. Pruned backward is not claimed. Exact and approximate workloads must not be mixed in a single speedup comparison without reporting numerical error.

6 Storage, memory and distribution

The storage hierarchy makes the capacity strategy explicit (Figure 3): keep the complete relation and source fields in a persistent store, stage one destination-row page through host memory, and execute with a bounded temporary GPU working set. The final output remains in device memory. FlashAttention’s IO-aware view distinguishes on-chip storage from device memory [5]; here the paging boundary also crosses host DRAM and NVMe. Tiga’s paging mechanism bounds graph residency across these outer memory levels; tile-local fusion bounds intermediate storage within a kernel.

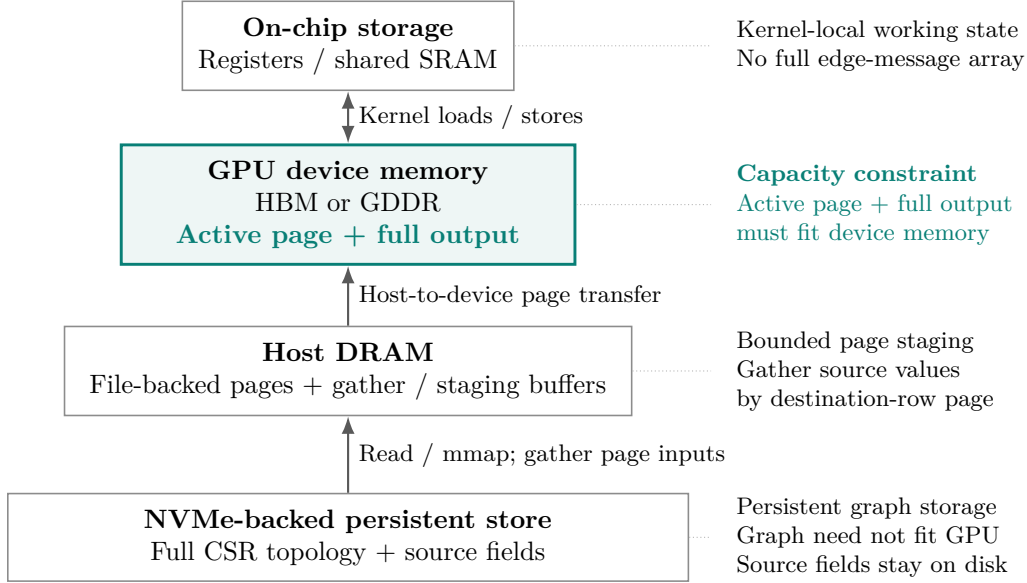


Figure 3: Storage hierarchy for paged graph execution. The central constraint is *active page plus full output*, not full-graph residency on the GPU. Arrows show input movement and kernel access; the implementation also returns each completed output page through host bytes before writing it into the resident device output. No direct NVMe-to-GPU path is implied. Widths are schematic, not hardware capacities or bandwidths. Experimental hardware and measured footprints are reported in Section 7.1 and Section 7.3, respectively.

6.1 Logical values versus physical residence

The native runtime records value metadata and physical buffers separately. Execution scopes select default device, tier budgets, spill directory and graph paging defaults; they do not migrate preexisting values. Memory reports count scope-created allocations, not process RSS or Torch-owned storage. Limits are budgets, not reservations of physical capacity.

Ordinary Torch storage remains under Torch’s allocator. Applications that need TIGA-managed paging or distributed ownership use native `tg.Tensor` fields for those paths, keeping the same message/reducer definition. The memory API and paged and distributed examples show the allocation scopes, graph snapshots and rank setup used by this runtime.

Temporary spill releases the tensor’s resident-buffer reference while retaining shape, dtype, logical device, version and same-process gradient history. Observation restores the value. Other views or launches may still own storage, so logical eviction is not proof that all associated device bytes were freed. Persistent snapshots store values, not a reconstructible autograd graph. Device copy is differentiable on supported native paths but currently eager, not merely a scheduler hint. The optional LRU policy evicts whole idle native tensors; it does not tile a working set larger than device memory, manage Torch allocations or supply transparent GPU-to-NVMe execution for arbitrary kernels.

Paged CSR streams bounded destination-row ranges on CPU and, for the tested FP32 forward slice, on CUDA via host staging. CPU paging has a VJP path; CUDA paged backward remains unsupported. Graph offload and field storage have distinct memory lifetimes; automatically paging topology does not imply every node/edge field is paged. Forward output assembly allocates the final resident tensor once and writes each completed page into its row range. Only one output-page payload is retained on the host; the final output must still fit the selected device. Persistent CSR ingestion in the large-graph experiment writes bounded chunks directly to the versioned store, without constructing a graph-sized Python index list.

This bounded-memory path is currently a capacity mechanism rather than an optimal IO pipeline. Source fields are gathered on the host into page-edge order and copied to CUDA, so

repeated source IDs can duplicate feature traffic within a page. Each completed output page is copied back to host bytes before being written into the final device output. Page-local staging bounds the temporary working set but does not eliminate these copies. Section 7.3 measures their cost and separates it from GPU kernel execution. Deduplicating source staging, direct device output assembly and asynchronous IO are distinct optimizations; they are not implied by an offload interface.

The optional runtime prefetcher maintains a bounded queue of topology-page reads and contiguous-field read-ahead hints. It does not move source gathers, host packing or CUDA copies into a compiler-scheduled asynchronous pipeline. Consequently prefetch depth is not a guarantee of disk/PCIe/kernel overlap; the capacity experiment reports the explicit serialized configuration.

6.2 Ownership, halos and progress

A rank is a cooperating worker process. A logical device mesh and destination partition assign output ownership to ranks. Remote source values needed by local destinations are held as ghost copies; the halo map specifies which values must cross rank boundaries. The public schedule completes halo exchange before computing all owned destination rows in one local call, without split-output assembly. Reverse-mode execution accumulates ghost contributions back to owners. Event ordering is a correctness requirement independently of whether communication and computation actually overlap in time. For example, if a destination on rank 0 reads a source owned by rank 1, the forward exchange copies that source value to rank 0. Backward sends the source’s gradient contribution in the opposite direction and adds it to the gradient on rank 1. This is why ownership must be retained through differentiation, not treated solely as a forward communication detail.

The runtime supports CPU rank-local execution, MPI transport, and two-host CUDA forward/VJP through TCP and NCCL. Deployment supplies worker processes, rank membership and a communicator. Ownership remains fixed during execution; Section 7.4 evaluates an equal-destination partition on spatial meshes.

6.3 Validation and trust boundaries

Dispatch distinguishes supported compiled paths from reference/provider routes. Full CSR validation checks monotonic row pointers, endpoints and source-index bounds; basic validation checks metadata and is not an exhaustive scan. Paged reads validate each consumed range. Tensor snapshots, geometry importers and TCP transports are not security sandboxes: compiled UDFs and providers execute trusted native code, and transport deployment requires trusted peers.

7 Evaluation: speed, capacity, and distributed execution

The evaluation measures runtime performance and allocated memory against matched PyTorch [17] and PyTorch Geometric (PyG) [6, 7] implementations, single-GPU capacity with storage offload, and the latency of distributed graph execution. Appendix A provides a differentiable EdgeNN example and a Torch forward/gradient reference. Numerical gradient validation is distinct from the forward-performance measurements reported here.

7.1 Setup and comparison boundaries

The single-GPU host has a Ryzen 7 255, approximately 44 GiB RAM, local NVMe storage and an RTX 5070 Ti with 16 GiB device memory (driver 595.84). The Torch path uses Torch 2.11.0+cu128, Triton 3.6.0, PyG 2.8.0.post1 and pyg-lib 0.9.0+pt211cu128. The second host supplies an RTX 4070 Ti SUPER with 16 GiB under Ubuntu/WSL. Existing services remain running, occupying roughly 3 GiB locally and 12 GiB remotely. These are shared hosts, not dedicated, clock-locked machines. GPU memory here means device memory; these consumer

GPUs use GDDR. Distributed communication uses NCCL 2.28.9 with its Socket transport across the two hosts. The reproduction archive specifies the network configuration.

Inputs, dimensions, precision, outputs and graph semantics are matched within each comparison. The primary comparisons are FP32 forward evaluations; Section 7.5 additionally measures differentiated EdgeNN steps. Performance comparisons use ordinary Torch tensors and public Tiga calls; paging and distributed experiments use native tensors. Timers cover synchronized host wall time through completed output, with numerical validation outside the timer. First compilation, warm execution and profiling have separate scopes. Speedup is

$$S = \frac{T_{\text{matched baseline}}}{T_{\text{Tiga}}}.$$

Values below one indicate slowdown. Source/compiler hashes, commands, versions and raw samples accompany each experiment’s source snapshot.

7.2 Runtime performance and memory efficiency

Three workloads isolate different uses of the graph abstraction:

- **Stored CSR:** degree 32, 32 features, weighted neighbor sum; 8,192, 32,768 and 131,072 nodes. Peers are `torch.sparse.mm` and PyG `MessagePassing` with COO gather-scatter. CSR setup is untimed; PyG’s required COO representation is included in its memory. The sparse peer prevents one COO path from standing in for all optimized PyG execution.
- **Dynamic radius:** scalar distance-weighted sum, nominal degree 32; 8,192, 32,768 and 131,072 points. Every call copies one of two different coordinate snapshots, rebuilds the relation and consumes it. Peers are PyG’s radius builder plus COO message passing and pure Torch chunked-distance CSR construction plus Torch sparse multiplication. The generated path avoids accepted-edge, distance and edge-message arrays.
- **Dynamic exact kNN:** scalar neighbor sum, $k = 16$; 1,024, 4,096 and 16,384 points, extended to 32,768, 65,536 and 131,072 points. The practical peer is PyG’s kNN builder plus COO message passing. The pure Torch peer computes direct squared distances in row chunks, selects top-k, then gathers and sums features. Neither Torch baseline calls a Tiga graph builder or uses approximate search.

Both dynamic snapshots undergo complete edge-set comparison between the explicit reference and PyG before timing. Radius coordinates use a binary grid and a cutoff between squared-distance levels to avoid ambiguous FP32 boundaries. The radius neighbor cap is 128; the complete comparison establishes that it truncates neither snapshot. kNN reference distances use direct differences, not the matrix-product identity. Every timed output is checked at relative and absolute tolerance $3 \cdot 10^{-4}$. Coordinates actually change; unchanged-input cache lookup is not counted as rebuilding. These generated dynamic paths use scalar features, not wide-feature edge networks.

The two Torch graph builders bound each distance block to 16,777,216 pairs (64 MiB for one FP32 distance plane; multiple temporary planes may coexist). They still inspect all candidate pairs and materialize the complete accepted relation. At 131,072 points an unchunked distance plane alone would require 64 GiB; chunking permits a real measured baseline instead of a predicted OOM. This is an eager all-pairs implementation, not an optimized spatial index or a claim that Torch cannot express a better algorithm.

Each of 36 configurations runs in a fresh process, with provider order randomized across the matrix, four warmups and ten checked timed calls. Curves show median and sample range, not cross-process confidence intervals. Peak Torch-allocated GPU bytes include live inputs, coordinate fixtures, output, provider-specific representations and temporary tensors. They exclude allocator reserve, CUDA context/module storage and other processes. Counters are reset per measured

call after warmup; reserved-memory peaks are not substituted for allocation. Separate largest-shape audits found zero additional TIGA-native device-buffer allocations on each of the three Torch paths.

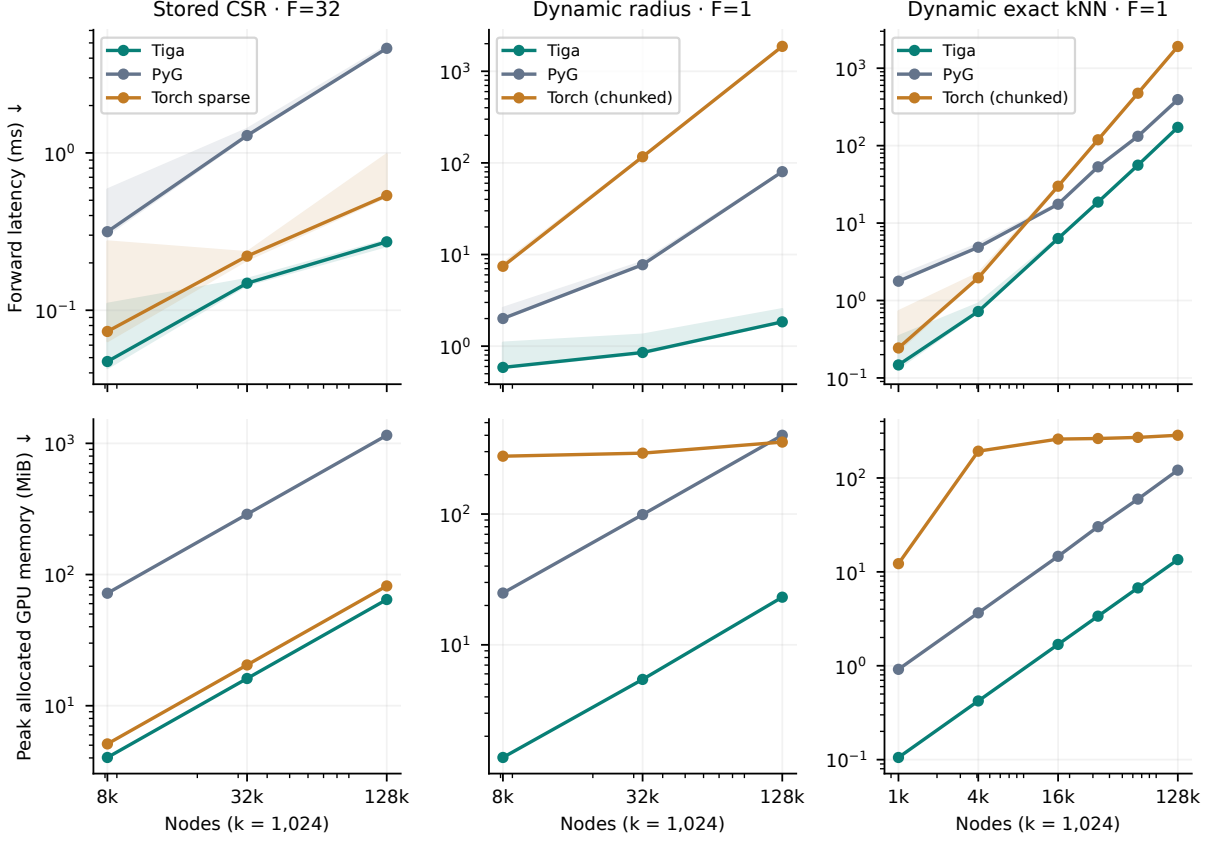


Figure 4: Matched forward latency and measured allocated GPU memory. Columns are different workloads; feature widths and sizes are not pooled. Both rows use logarithmic axes and lower is better. Dynamic calls rebuild and consume changed coordinates. Torch (chunked) means a pure Torch graph builder and Torch aggregation, not a TIGA-built graph. kNN extends through 128K points; the selected graph at that size has approximately 2.10M directed edges.

Stored CSR takes approximately 0.047–0.272 ms, giving 1.49 – $1.97\times$ speedup over Torch sparse on the three measured sizes. Allocation is about $1.27\times$ lower than the sparse peer, rather than the much larger reduction against materialized COO messages. The largest dynamic radius case takes 1.84 ms versus PyG’s 80.24 ms and pure Torch’s 1,870.09 ms; allocation peaks are 23.14, 400.22 and 355.01 MiB respectively. These are workload-specific outcomes, not a framework ranking.

Exact kNN uses semantic executable rebinding and the exact tile-selection pruning described in the compiler section. At 1,024 points TIGA takes 0.147 ms versus 0.244 ms for pure Torch and 1.775 ms for PyG. At 16,384 points the times are 6.32, 29.92 and 17.50 ms, respectively. At 131,072 points they are 172.37, 1,908.43 and 393.21 ms: TIGA is $2.28\times$ faster than the PyG peer with 13.50 versus 121.13 MiB allocated (pure Torch: 285.00 MiB). Across the six measured sizes the PyG-relative speedup is 2.28 – $12.06\times$. All paths rebuild relations from changed coordinates; executable reuse does not reuse selected neighbors. Candidate-distance work remains quadratic, so these results do not establish subquadratic search or a universal advantage over spatial-index kNN implementations.

7.3 Billion-edge execution on a single GPU

The capacity workload is an explicit periodic directed CSR graph with 16 neighbors per destination and 32 FP32 features. Every edge is persisted and traversed; the periodic closed-form oracle is used only for validation. The sweep stops at 10M, 100M and 1B edges by design, not at a demonstrated maximum supported size. Three fresh processes per size are retained from the fully checked capacity run. Bounded ingest is untimed; forward includes page loading, host staging, JIT as encountered, computation and output assembly, but excludes full-output checking.

Pages contain 65,536 destination rows, prefetch is disabled, and the native device budget is 12 GiB. This is a serialized capacity baseline, not a measurement of overlapped IO or compiler-scheduled prefetch. The runtime’s optional read-ahead covers topology loading and contiguous field hints; source gathering and GPU staging remain on the consuming thread. Enabling that option alone does not create a disk–host–device double-buffered pipeline. The final output remains resident. At 1B edges, 62.5M nodes require 8.5 GB of indices, 8 GB of source features and 8 GB of output. The full-residency lower bound is 22.82 GiB with int64 CSR, or 18.86 GiB with int32 CSR, both beyond 16 GiB. These are arithmetic infeasibility bounds, not measured OOM attempts or Torch/PyG runs at this size. A hand-chunked peer could also exchange storage traffic for capacity; paging is not claimed to be unique.

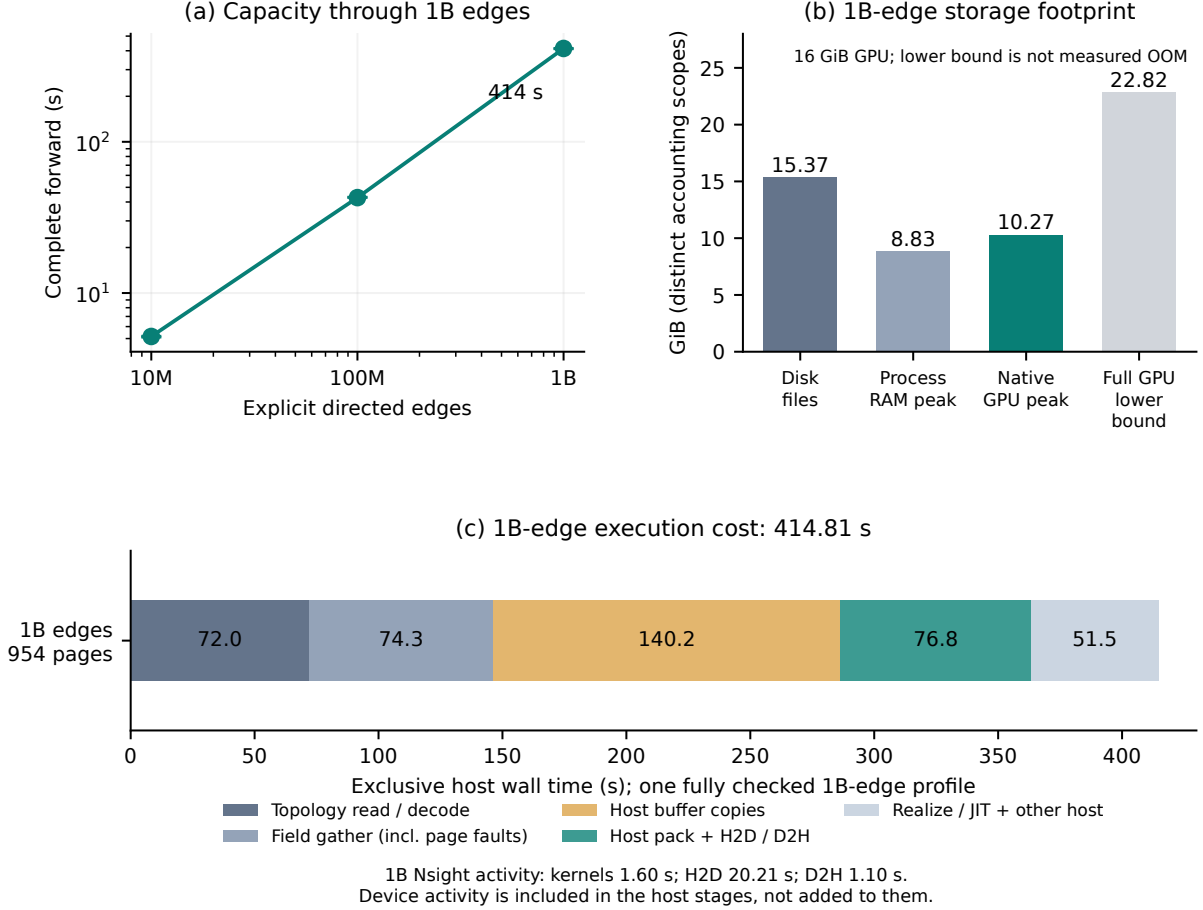


Figure 5: Single-GPU capacity and execution cost. (a) Median/range of three complete calls per size. (b) Disk footprint, process RAM peak, native GPU peak and calculated full-residency requirement at 1B, with distinct accounting scopes. (c) Exclusive host stages in one fully checked, instrumented 1B-edge forward call. Nsight CUDA activity is already contained in the host intervals, not an extra stage.

The 1B median is 414.25 seconds (411.36–415.61), with maximum native GPU peak 10.27 GiB

and forward process RSS 8.83 GiB. All two billion output scalars are checked exactly. Topology and source occupy 15.37 GiB on disk. Device-wide sampled usage, including services and allocation pools, reaches 13.25 GiB; the native counter alone is not total physical usage. The output occupies 7.45 GiB, so capacity still depends on node count and feature width.

A separate instrumented 1B-edge call records exclusive nested host intervals and an Nsight CUDA capture restricted to forward. All two billion output scalars are checked exactly afterward. The profiled window takes 414.81 seconds; the outer call takes 417.05 seconds including profiler start/stop overhead. It is not pooled into the original three-trial latency sweep. Topology read/decode takes 71.98 seconds, field gathering 74.27 seconds, host-buffer copying 140.22 seconds, packing and host/device transfer stages 76.79 seconds, and realization/setup/assembly 51.55 seconds. These exclusive host intervals sum to the profiled window. Nsight separately records 1.60 seconds of CUDA kernels, 20.21 seconds H2D and 1.10 seconds D2H; these device activities are already contained in the host stages.

Host staging, rather than kernel arithmetic, dominates. Source staging expands feature rows at edges: an 8 GB stored source field generates 128 GB of gathered values. Total H2D traffic is 144.50 GB; D2H is 8 GB. Eviction hints on only the three fixture files yield 16.50 GB of process physical reads, including any runtime/compiler file reads. mmap gathering includes page-fault IO, not an isolated disk-service timer. The graph fits host RAM; this is not evidence for a graph beyond RAM capacity. CUDA paged backward, 1B recurrence and output offload remain outside the demonstrated result.

Page-size sensitivity. A supplementary 10M-edge sweep holds the graph, 32-feature operation and 12 GiB budget constant while varying destination rows per page. Nine fresh processes (three per setting, randomized order) run the same public paged path with prefetch disabled. Each checks all 20 million output scalars exactly. Forward includes JIT as encountered; ingestion and validation are excluded. The OS page cache is not flushed, so this is not a cold-NVMe measurement. Larger pages increase peak allocation without monotonically improving time:

Rows per page	Pages	Forward (s)	Native peak (MiB)
16,384	39	4.95 (4.94–4.96)	435.5
65,536	10	4.87 (4.81–4.90)	935.3
262,144	3	5.41 (5.35–5.46)	1442.1

Table 6: Page-size sensitivity at 10M edges. Forward shows median and range of three complete calls; native peak is the maximum scoped device allocation. It excludes CUDA context, external services and OS page cache.

65,536 rows are slightly faster here than either alternative. Fixed setup and host-staging costs limit gains from reducing page count. This small-graph study does not identify an optimal 1B schedule, establish capacity beyond host RAM, or compare against another bounded-memory implementation.

7.4 Distributed execution and communication overhead

The 5070 Ti and 4070 Ti SUPER each own half the destination rows of one graph, exchanging remote source features through halo maps. This is two-rank graph partitioning, not feature-axis tensor parallelism or independent data-parallel graphs. Ownership is fixed for each run.

The workload is a nonperiodic structured three-dimensional hexahedral mesh. Nodes sharing an element are connected in both directions, excluding self entries: an interior node has 26 neighbors, and exterior nodes have fewer. This matches the connectivity of a trilinear hexahedral FEM discretization; the measured operation is 16-feature neighbor summation, not stiffness assembly or a complete FEM solve. Cubes have side lengths 32, 64 and 96, giving 32,768, 262,144 and 884,736 nodes and 797,816, 6,596,856 and 22,508,920 directed edges.

Lexicographic x-major numbering makes the equal-row partition a spatial plane. Each rank owns half the volume and exchanges one interface layer rather than a volume-sized ghost set. For a cube of side length L ,

$$N = L^3, \quad N_{\text{halo,rank}} = L^2, \quad f_{\text{boundary}} = \frac{2}{L}.$$

The boundary fractions decrease from 6.25% to 3.125% and 2.083%; total traffic across both ranks is 0.125, 0.500 and 1.125 MiB per call. Thus the experiment measures the small surface-to-volume regime relevant to spatial graphs.

Each GPU also executes the complete graph alone. The *communication then compute* schedule completes halo exchange before computing local outputs. Two warmups precede five changed-input, fully checked calls. Per-sample distributed latency is the maximum of paired rank durations, followed by a median, not the mean of rank medians. Topology setup, input allocation, the control barrier and checking are excluded. Setup topology is replicated and partition maps cached: this is strong scaling, not distributed topology-capacity evidence beyond one GPU.

Both hosts use matching Tiga sources and compiler executables. Single-card and two-card modes share the native-input timing boundary; their times are not directly comparable with the different ordinary-Torch workloads in Section 7.2. Modes run in fixed order; small sample counts and shared services limit cross-run generalization.

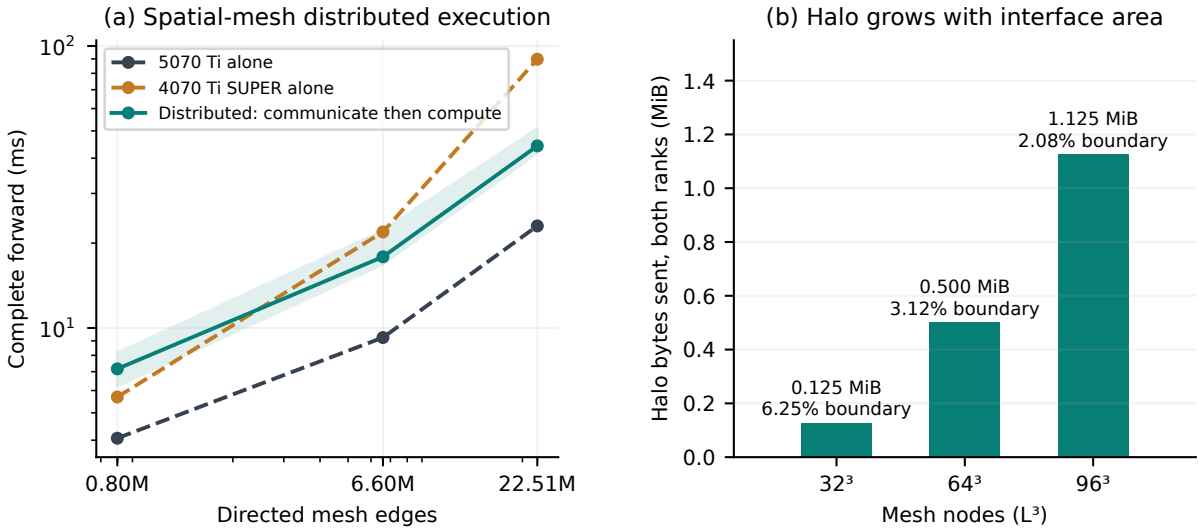


Figure 6: Distributed NCCL execution of spatial mesh graphs on two hosts. (a) Full-graph single-GPU baselines and paired two-rank latency. (b) Halo traffic grows with the interface area while the boundary fraction shrinks with mesh size. Latency covers the complete forward call: halo exchange, runtime scheduling, computation and synchronization. Each rank owns half the destination nodes.

At 22.51M directed edges, the 5070 Ti and 4070 Ti SUPER single-GPU medians are 23.00 and 89.75 ms. The communication-then-compute schedule takes 44.22 ms on the paired-rank critical path, 1.92 times the faster single-GPU latency. Only 2.08% of each rank’s destination rows touch the interface, and total halo traffic is 1.125 MiB. This fixed partition assigns half the destination rows to each GPU. Compute-aware partitioning is a potential optimization for the measured heterogeneous pair.

7.5 Controlled execution studies

The following studies distinguish stored-topology execution, differentiated message computation and first-call cost from the dynamic-search comparisons. They use the same single-GPU host,

with three fresh processes per configuration, seeds 20260921–20260923, four warmups and ten checked calls per process. Configuration order is randomized. Plots show the median and range of process medians, not a confidence interval. Feature values alternate between two snapshots; the input copy and checking are outside the synchronized wall timer. Reported memory is peak Torch-allocated storage, including live topology, inputs, outputs, tapes and workspaces, but excluding allocator reserve, CUDA context and other processes. The source snapshot is independent of the older experiments; unchanged hardware does not imply an unchanged implementation.

Holding topology constant. A common radius search produces a fixed CSR snapshot at 8,192 or 32,768 uniform binary-grid points, with nominal degree 32. The three consumers receive identical indices, Euclidean weights and 32- or 128-component features: Tiga’s compiled weighted sum, Torch gather–multiply–scatter, and Torch sparse matrix multiplication. Search is outside timing for all three. CSR hashes are checked between peers in each trial. This is a topology-controlled execution-path comparison, not a single-pass on/off ablation: traversal, dispatch and reduction strategy can still differ.

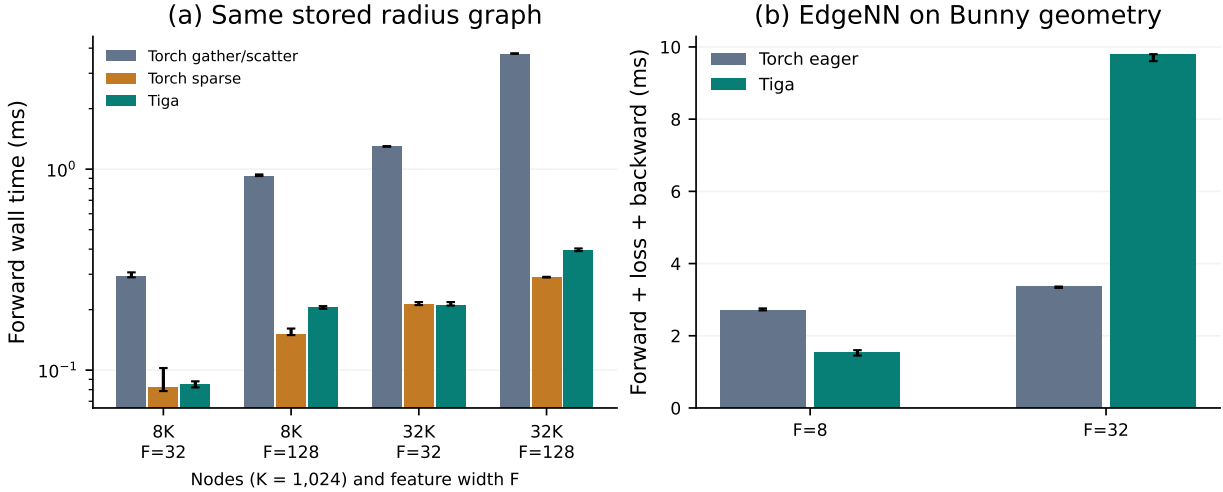


Figure 7: Supplementary execution studies. (a) The same stored radius topology is consumed by three implementations; graph construction is excluded. (b) Differentiated EdgeNN steps on a radius graph of Stanford Bunny geometry. Bars show medians of three process medians, with their ranges. Lower is better; only panel (a) uses a logarithmic axis.

At 32,768 nodes and 128 features, measured forward times are 0.395 ms, 3.759 ms, 0.289 ms for Tiga, Torch gather/scatter and Torch sparse, respectively. Median process-peak allocations are 67.3 MiB, 1007.3 MiB, 67.6 MiB. These measurements hold search and topology fixed; execution strategy and launch overhead still differ. The reduction in intermediate storage is clear against explicit message materialization; the optimized sparse-library peer can be faster than Tiga and has similar storage. Thus eliminating edge messages is useful but does not alone establish a better schedule than a sparse library. This study does not isolate the separate gain from avoiding accepted-edge storage in the generated-radius experiment.

Differentiation on real geometry. The Stanford Bunny zippered reconstruction provides 35,947 reconstructed vertices [19]. Coordinates are centered and divided by the largest bounding-box extent; radius 0.015 gives 356,260 directed non-self edges. The reconstructed surface is the data source, not raw scanner noise. Features are synthetic; an untrained MLP maps displacement and source features through a 32-unit ReLU layer to three outputs, summed at destinations. The mean squared difference from normalized positions supplies a scalar regression loss. Both paths use the same fixed graph and parameters. This workload measures a differentiated geometric

operator, not downstream task accuracy, convergence or a complete optimizer step.

Every retained call checks outputs and gradients of features, positions and all four parameter tensors against explicit Torch gather–MLP–scatter. Position gradients hold edge membership fixed. In addition to elementwise tolerances, each gradient must have relative L2 error below 0.002. The Tiga path must report its compiled tile VJP; a reference fallback is rejected. Table 7 separates forward from loss-plus-backward. Independent medians of these stages need not sum exactly to the median total.

Input width	Path	Forward (ms)	Backward (ms)	Total (ms)	Peak (MiB)
8	Torch eager	0.64	2.07	2.71	172.25
8	Tiga	0.86	0.72	1.54	32.34
32	Torch eager	0.94	2.42	3.36	209.22
32	Tiga	0.92	8.86	9.79	37.58

Table 7: Bunny EdgeNN: medians across three process medians. Peak allocation is the median of per-process measured peaks. Backward includes loss construction; no optimizer update is timed.

At width eight, the measured total improves from 2.71 to 1.54 ms with substantially less allocated storage. At width 32, Tiga uses less storage but its 8.86 ms backward makes the total slower than Torch. Recompute-based fusion is therefore a memory–time tradeoff, not a universal training acceleration. This experiment does not independently isolate recomputation, atomic accumulation and scheduling as causes of that slowdown.

First result and amortization. For the 32,768-node, 32-feature stored-radius case, three additional worker pairs use empty isolated Triton, TorchInductor and CUDA caches, then reuse each pair’s disk caches in a new process. Imports, CUDA initialization, graph setup and the eager correctness oracle precede timing. The first result includes Tiga capture, lowering, provider work, module loading and execution, rather than pure compile time. A disk-cache hit does not restore an in-process executable. The gather/scatter peer has already served as the correctness oracle, so its entry is the first timed call, not cold library initialization. All entries exclude that common setup; this is an execution-ready comparison rather than time from launching Python to a first answer.

Path	First result (ms)	Warm call (ms)
Tiga, empty isolated caches	1105.5 (1095.5–1113.8)	0.214
Tiga, reused disk caches	635.1 (621.9–651.4)	0.216
Torch sparse, fresh process	17.0 (16.9–17.5)	0.213
Torch gather/scatter, fresh process	1.3 (1.2–1.3)	1.301

Table 8: First-result cost versus warm execution. First-result entries show median and range of three fresh processes; warm entries are medians of process medians. Common setup is excluded; the gather/scatter oracle is already warm. Torch sparse uses its ordinary library environment.

For first-result cost C , warm time t and K calls, an amortization model is

$$T(K) = C + (K - 1)t.$$

The model assumes stable warm time; it is not a measured cumulative-time curve. Here the cold first result costs about 1.11 s, and persistent-cache reuse still costs about 0.64 s. Warm Tiga and Torch sparse are both about 0.21 ms, with no measured steady-state advantage supporting an amortized crossover against the sparse peer. Against explicit gather/scatter, warm savings can amortize setup over repeated calls. Using the reported medians, the model predicts a crossover against explicit Torch gather/scatter after approximately 1,018 calls. This is a model-derived

estimate, not an observed crossover. Short-lived programs should not be assigned warm-kernel speedups without this qualification.

7.6 Reproduction and scope

`data/comparison-v3/` stores the 36 comparison records, source snapshot and allocation audits. `data/billion/` contains the nine capacity trials; `data/paging-1b/` contains the instrumented billion-edge run and Nsight capture. `data/distributed-spatial/` contains both rank records, commands, source and interface-size checks for the spatial-mesh experiment. `tools/build_three_questions.py` validates the archived configurations and numerical checks before plotting. Rerunning hardware measurements uses the commands and environments provided with each dataset. `data/supplement/` adds the controlled execution and first-call records, with `tools/run_supplement.py` and `tools/build_supplement.py` for measurement and validation. The Stanford archive URL and checksum identify the external geometry input; its model is not required to compile this report. `data/page-sensitivity/` retains the nine page-size trials and commands; `tools/build_page_sensitivity.py` validates and summarizes them.

The measurements support three conclusions: generated relations can reduce runtime and intermediate storage; paging permits billion-edge execution beyond device-memory capacity; and communication plus heterogeneous load balance determines the cost of distributed execution. Gradient correctness is covered separately by the differentiable examples and path-specific checks. End-to-end model-training throughput, capacity beyond the measured edge count, and scaling to larger device meshes remain subjects for further evaluation.

8 Limitations and next steps

The implementation has the following scope limits:

- Native and Torch paths have different operator, dtype and gradient coverage. Traced modules support a guarded subset; structured reverse loops and general higher-order differentiation remain incomplete.
- Generated exact kNN supports a bounded FP32, small-k envelope. Larger-k scratch, general metric lowering and generated backward require further work.
- Implicit block composition, automatic heterogeneous out-of-core execution and additional vendor backends remain incomplete.
- Distributed execution uses fixed ownership. Compute-aware partitioning, partition-local topology capacity, RCCL and failure recovery remain open.
- Performance evaluation covers FP32 forward workloads and a fixed-topology EdgeNN forward/backward step, not end-to-end model training. The latter shows a memory reduction but a wide-feature backward slowdown. Sustained out-of-core training and beyond-host-RAM graphs require additional measurement.

9 Conclusion

Tiga compiles graph message-passing programs while retaining the relations, reducer algebra and derivative requirements that determine their execution. This separation supports CPU and GPU lowering, forward and backward computation on supported paths, and explicit control over storage residence and distributed ownership. Generated relations reduce intermediate storage and improve runtime on the measured workloads; paging enables billion-edge forward computation on a memory-limited GPU. Distributed measurements show that a small halo alone is insufficient for speedup in the measured heterogeneous deployment. Device speed, host environment and competing services are not independently controlled, so the observed gap does not isolate their individual effects.

The design links a local programming abstraction to the physical choices that govern large-graph execution. Further work can build on the same representation to extend backward coverage, reduce staging traffic and balance computation across heterogeneous devices.

Reproducibility and tool use

The report repository provides measurement records, numerical checks, source snapshots, and figure-generation scripts. Each experiment identifies its timing and allocation boundary; historical runs retain their original source snapshot. Replotting recorded samples does not constitute a new hardware measurement. The supplementary experiments include explicit compiler-path checks, and their inputs and gradients are checked independently of the measurement window.

AI coding and writing assistants were used in software implementation and testing, experiment design and tooling, literature discovery, interpretation of measurements, figure/code preparation, and manuscript drafting and revision. Numerical results originate from recorded program executions, not generated estimates. Automated numerical and provenance checks were used to validate outputs, supported gradients and the recorded evidence. The human author is responsible for the final content, attribution and claims.

A A differentiable EdgeNN example

This example expresses a shared edge-local neural network followed by a destination-wise sum. Positions and node features are ordinary `torch.Tensor` objects; the weights are ordinary `torch.nn.Parameter` objects. No native tensor wrapper or handwritten backward is required. For a radius relation, the program computes

$$\mathcal{N}(i) = \{j \neq i : \|p_j - p_i\|_2 \leq r\}, \quad y_i = \sum_{j \in \mathcal{N}(i)} [W_2 \text{ReLU}(W_1[p_j - p_i; x_j] + b_1) + b_2].$$

Here the final bias is part of each edge message, inside the sum. The feature widths are 8 for node inputs, 3 for displacement, 16 for the hidden layer and 4 for each output. The reducer produces a zero vector for a destination with no neighbors.

A.1 Define the message and differentiate the result

`tg.nn.trace` captures the supported Torch module without replacing its parameters. Its multiple tensor arguments are concatenated along the feature axis: `edge.displacement` supplies the first three features and `src.x` the remaining eight. `reducer = tg.sum()` specifies aggregation once on the class, rather than inside the edge function.

```
import torch
from torch import nn
import tiga as tg

torch.manual_seed(7)
device = "cuda" if torch.cuda.is_available() else "cpu"
positions = torch.rand(128, 3, device=device, requires_grad=True)
x = torch.randn(128, 8, device=device, requires_grad=True)
mlp = nn.Sequential(nn.Linear(11, 16), nn.ReLU(), nn.Linear(16, 4)).to(device)

class EdgeNN(tg.MessagePassing):
    reducer = tg.sum()

    def __init__(self, module):
        super().__init__()
        self.mlp = tg.nn.trace(module)

    def edge(self, src, dst, edge):
        return self.mlp(edge.displacement, src.x)

graph = tg.Graph.radius(positions, cutoff=0.35)
program = EdgeNN(mlp)
y = program(graph=graph, src={"x": x}, dst={})
assert y.shape == (128, 4)
loss = y.square().mean()
variables = (x, positions, *mlp.parameters())
grads = torch.autograd.grad(loss, variables)
```

The call returns one four-component vector per destination. The loss yields gradients for node features, positions and every MLP weight and bias through `torch.autograd.grad`; `loss.backward()` is the usual alternative when accumulated `.grad` fields are wanted. The position derivative acts through the displacement of selected edges. Discrete radius membership is not differentiated: moving across the cutoff changes the selected relation.

A.2 Check against an explicit Torch reference

The following continuation materializes edges only for validation, applies the original Torch MLP, and aggregates with `index_add`. Both forward values and all gradients are compared on the same selected relation. This isolates message, reducer and differentiation correctness; it is not an independent validation of the radius-neighbor search.

```
# Independent gather -> MLP -> index_add reference on the same relation.
row_ptr, col_idx = graph.resolve_csr()
rows = torch.repeat_interleave(
    torch.arange(128, device=device), row_ptr[1:] - row_ptr[:-1])
delta = positions[col_idx] - positions[rows]
messages = mlp(torch.cat((delta, x[col_idx]), dim=-1))
reference = torch.zeros_like(y).index_add(0, rows, messages)
reference_grads = torch.autograd.grad(reference.square().mean(), variables)
torch.testing.assert_close(y, reference, rtol=2e-4, atol=2e-5)
for actual, expected in zip(grads, reference_grads):
    torch.testing.assert_close(actual, expected, rtol=3e-4, atol=3e-5)
print(f"EdgeNN forward and all gradients checked on {device}.")
```

On the supported CUDA EdgeNN path, Tiga fuses edge evaluation and reduction into tiled execution and recomputes local activations in the VJP, avoiding a global edge-by-hidden-feature activation tensor. CPU execution remains useful for functional checking but does not establish that CUDA performance property. Unsupported module operations are outside this example's capture contract. The complete runnable file is `examples/edge_nn.py`; execute it from the paper directory with Tiga and the appropriate Torch installation:

```
python examples/edge_nn.py
```

B Causal attention as message passing

Causal attention uses the same interface as sparse aggregation: the graph selects admissible interactions, the edge function computes a score and value, and the reducer normalizes their weighted sum. No attention-specific graph operator is required. For each head, the computation is

$$\mathcal{N}(i) = \{0, \dots, i\}, \quad s_{ij} = \frac{q_i^\top k_j}{\sqrt{D}}, \quad y_i = \frac{\sum_{j \in \mathcal{N}(i)} e^{s_{ij}} v_j}{\sum_{j \in \mathcal{N}(i)} e^{s_{ij}}}.$$

The destination supplies a query; each source supplies a key and value. The triangular relation includes the diagonal: token zero reads only itself; token one reads tokens zero and one. Causality therefore belongs to the relation, not to an extra condition inside the edge function.

B.1 Define and execute the program

`tg.online_softmax()` declares the reducer once. Calling `self.reducer(score, value)` packages its two inputs; it does not run a separate softmax for each edge. The compiler combines these messages over each destination’s neighbors using the stable state in Section 5. Pruning is disabled by default, so this example evaluates the full causal relation, subject to floating-point rounding.

```
import torch
import tiga as tg

class CausalAttention(tg.MessagePassing):
    reducer = tg.online_softmax()

    def edge(self, src, dst, edge, scale):
        score = (dst.query * src.key).sum(dim=-1) * scale
        return self.reducer(score, src.value)
```

Inputs and outputs are ordinary Torch tensors with logical shape (N, H, D): sequence length, number of heads and head width. The CUDA streaming implementation currently expects head-major storage, obtained by viewing a contiguous (H, N, D) allocation through `permute`. This changes strides, not the public dimension order, and does not copy the data. The runnable check defaults to `device="cuda"` and `nodes=65`, with two heads of width 64. The sequence length exercises a partial tile.

```
torch.manual_seed(7)
heads, width = 2, 64
dtype = torch.float16 if device == "cuda" else torch.float32
# Head-major storage, viewed as ordinary (N, H, D) Torch tensors.
query, key, value = (
    torch.randn(heads, nodes, width, device=device, dtype=dtype)
    .permute(1, 0, 2) for _ in range(3))
graph = tg.Graph.triangular(nodes, device=device)
program = CausalAttention()
with torch.no_grad():
    output = program(graph=graph, src={"key": key, "value": value},
                     dst={"query": query}, scale=width**-0.5)
```

On CUDA, the script checks the lowering reported by `program.explain()`. It requires the compiled dense streaming path. This forward traverses the implicit triangular relation without allocating a global pairwise score or probability tensor. Changing the constructor to `tg.Graph.dense(N)` expresses unmasked attention with the same edge function and reducer.

B.2 Validate outputs and query/key/value gradients

An independent Torch implementation explicitly forms scores, masks future positions, applies softmax and multiplies by values. This dense implementation is a small correctness oracle, not the memory-efficient execution path.

```
def torch_attention(query, key, value):
    # Public shape (N, H, D); matrix multiplication uses (H, N, D).
    q, k, v = (x.transpose(0, 1) for x in (query, key, value))
    scores = (q @ k.transpose(-1, -2)) * q.shape[-1]**-0.5
    future = torch.ones(q.shape[1], q.shape[1], device=q.device,
                        dtype=torch.bool).triu(1)
    weights = scores.masked_fill(future, -torch.inf).softmax(-1)
    return (weights @ v).transpose(0, 1)
```

The script compares the FP16 CUDA streaming output with this FP32 oracle. The current streaming kernel is forward-only; setting `requires_grad` does not supply a compiled attention backward. For a separate first-order derivative check, `program.reference` explicitly executes the differentiable Torch semantics in FP32. Independent leaf tensors and a random output cotangent test all three input gradients, rather than just a summed-output loss.

```
variables = tuple(x.float().detach().requires_grad_()
                  for x in (query, key, value))
q, k, v = variables
semantic = program.reference(
    graph=graph, src={"key": k, "value": v},
    dst={"query": q}, scale=width**-0.5)
reference_variables = tuple(x.detach().clone().requires_grad_()
                           for x in variables)
reference = torch_attention(*reference_variables)
cotangent = torch.randn_like(reference)
gradients = torch.autograd.grad(semantic, variables, cotangent)
expected_gradients = torch.autograd.grad(
    reference, reference_variables, cotangent)
torch.testing.assert_close(semantic, reference, rtol=2e-4, atol=2e-5)
for actual, expected in zip(gradients, expected_gradients):
    torch.testing.assert_close(actual, expected, rtol=3e-4, atol=3e-5)
```

The checks establish compiled-forward agreement and semantic gradient agreement separately. The reference path materializes the selected edges and their messages; it is intended here for small validation problems and does not establish streaming training memory or throughput. A fused attention VJP is a separate implementation requirement from the EdgeNN VJP in Appendix A.

The complete file, including imports and assertions, is `examples/causal_attention.py`. With Tiga, its native compiler tools, Torch and the CUDA provider installed, execute it from the paper directory:

```
python examples/causal_attention.py --device cuda
```

The optional `-device cpu` runs semantic checks only; `-nodes` changes the sequence length. The CUDA check fails if the streaming lowering is unavailable instead of accepting a reference fallback.

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